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Cli-fi in the language and literature classroom: impact of reading and argumentative writing

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Climate fiction (cli-fi) has emerged as a relevant pedagogical resource for addressing the climate crisis within language and literature education, particularly in relation to critical reading, argumentative writing, and climate literacy. This study aims to analyze the educational use of cli-fi and its reported impact on discursive competencies and environmental awareness. A systematic literature review (SLR) was conducted following the PRISMA 2020 protocol, using Scopus and Web of Science databases. From an initial corpus of 98 records, 57 studies published between 2015 and 2025 were selected. A combined bibliometric and qualitative synthesis was applied to identify thematic trends, pedagogical strategies, and educational outcomes. The results reveal a sustained growth in scientific production from 2020 onwards, with a predominance of studies in higher education and within the humanities. Three main pedagogical approaches were identified: critical reading and discussion, creative writing and speculative narratives, and climate education projects focused on climate justice and sustainable futures. The findings indicate positive effects on climate literacy, ecological awareness, student engagement, and socio-environmental agency, while evidence regarding improvements in argumentative writing remains limited and predominantly qualitative. Additionally, the literature is characterized by a prevalence of qualitative designs and small-scale case studies, with scarce experimental or mixed-methods research. In conclusion, cli-fi constitutes an emerging and expanding field with significant potential to integrate climate education and discursive development; however, further empirical research is required to strengthen evidence on its impact on argumentative competencies and to support its systematic incorporation into educational curricula.

KEYWORDS
argumentative writing, cli-fi, climate fiction, climate literacy, critical reading, literary education

1 Introduction

Climate fiction (cli-fi) has gained increasing relevance over the past two decades as a narrative resource that depicts scenarios of ecological crisis, speculative futures, and sociopolitical transformations associated with climate change. This type of narrative has been studied from literary, cultural, and ecocritical perspectives, highlighting its capacity to problematize tensions between nature, technology, and human agency in the face of planetary crisis (Salomon, 2024; Yonar, 2025). In this sense, cli-fi engages

with contemporary concerns linked to the Anthropocene and climatic imagination, offering a discursive space to interrogate the ways in which possible environmental futures and associated ethical frameworks are constructed (Kwintowski, 2025).

Various works have emphasized the affective and emotional dimension of cli-fi, which contributes to exploring climate anxieties, ecological grief, and hopes tied to socio-environmental resilience (Klosterman and Pérez, 2024). These approaches underline that cli-fi operates not only as literary representation of climate change, but also as cultural mediation that articulates affects, cognition, and ethics in contexts of environmental crisis. This articulation has enabled cli-fi to connect with debates on climate justice, sustainability, and the configuration of socioecological futures, consolidating itself as an expanding interdisciplinary field.

In recent years, academic interest in cli-fi has extended to the educational domain, particularly in relation to literary education and climate literacy. Several studies have examined the formative potential of cli-fi to promote reflective and critical processes regarding climate change, as well as to stimulate the imagination of sustainable futures and ecological sensitivity (Reynante et al., 2025). These works have helped position cli-fi as an emerging pedagogical resource capable of integrating aesthetic, affective, and cognitive dimensions in the formation of environmentally informed readers and citizens.

From the perspective of Language and Literature education, argumentative writing constitutes a central discursive competence associated with critical literacy, the construction of evidence-based claims, and ethical positioning in socio-scientific controversies. Argumentation involves not only the formal structuring of claims, reasons, and counterarguments, but also the ability to situate textual interpretations within broader socio-cultural and political frameworks. In the context of climate change education, argumentative writing plays a key role in enabling students to articulate informed positions on environmental issues, negotiate competing narratives, and critically evaluate dominant discourses surrounding sustainability and climate justice. In this regard, cli-fi can function as a narrative trigger that scaffolds argumentative processes by providing complex scenarios, moral dilemmas, and speculative futures that invite interpretative debate and reasoned positioning.

Nevertheless, despite growing academic interest, the educational use of cli-fi remains dispersed and poorly systematized. Most studies focus on literary or theoretical perspectives, while pedagogical, curricular, and didactic dimensions receive less attention. The relationship between cli-fi, critical reading, argumentative writing, and climate literacy constitutes an incipient field that requires more structured examination, especially in contemporary school and university contexts. Similarly, questions persist regarding the role of cli-fi in the Language and Literature curriculum and its contribution to the development of discursive competencies associated with argumentation.

In this context, the present study conducts a bibliometric analysis and a Systematic Literature Review on the use of cli-fi in education, with special attention to the development of critical reading, argumentative writing, and climate literacy competencies. The main contribution of the study lies in synthesizing the state of knowledge, identifying pedagogical strategies and reported educational outcomes, and pointing out gaps and opportunities for integrating cli-fi into the Language and Literature classroom within an educational horizon crossed by the climate crisis.

2 Methodology

2.1 Study design

The present work was developed following the approach of a **Systematic Literature Review (SLR)**, aimed at the identification, selection, evaluation, and synthesis of relevant scientific publications on the use of climate fiction (cli-fi) in educational contexts, particularly in the teaching of Language and Literature and in the development of reading and argumentative writing competencies. The guidelines of the **PRISMA 2020** protocol were adopted to ensure transparency, reproducibility, and methodological rigor throughout the entire process.

2.2 Research questions

The review was guided by the following questions:

RQ1: How has cli-fi been addressed in formal educational contexts?
RQ2: What pedagogical strategies or approaches have been reported for its integration into the Language and Literature classroom?
RQ3: What effects are documented on competencies of critical reading, argumentation, and academic writing?
RQ4: What gaps, limitations, and opportunities emerge for future research?

2.3 Information sources and databases

The documentary search was conducted on **January 9, 2026** in three international databases with broad coverage in the social sciences, humanities, and education:

- **Scopus** (Elsevier)
- **Web of Science Core Collection** (Clarivate Analytics)

These databases were selected for their disciplinary complementarity: Scopus and WoS provide greater bibliometric visibility and multidisciplinary coverage.

2.4 Search equation

To ensure consistency and comparability, structured search equations were used around three conceptual cores: (a) cli-fi/ climate fiction, (b) education/teaching, and (c) argumentative reading-writing/literacy.

The equations were syntactically adapted to each database. Example:

Scopus: TITLE-ABS-KEY("cli-fi" OR "climate fiction" OR "eco-fiction") AND TITLE-ABS-KEY("education" OR "teaching" OR "classroom" OR "language arts" OR "literature") AND TITLE-ABS-KEY("reading" OR "writing" OR "argumentative writing" OR "critical literacy")

WoS: TS = ("cli-fi" OR "climate fiction" OR "eco-fiction") AND TS = ("education" OR "classroom" OR "language arts" OR "literature") AND TS = ("reading" OR "writing" OR "argumentative writing" OR "critical literacy")

2.5 Inclusion and exclusion criteria

Inclusion Criteria (IC):

- **IC1:** Empirical, theoretical, or review studies and conference papers.
- **IC2:** Publications in academic journals or indexed books.
- **IC3:** Focus on cli-fi, climate fiction, or eco-fiction.
- **IC4:** Application in formal or non-formal educational contexts.
- **IC5:** Orientation toward Language, Literature, or literacy.
- **IC6:** Evaluation of critical reading, writing, or argumentation.
- **IC7:** Accepted languages: English, Spanish, Portuguese, and French.

Exclusion Criteria (EC):

- **EC1:** Works without an educational component.
- **EC2:** Purely literary studies without relation to teaching.
- **EC3:** Documents without access to the full text.
- **EC4:** Proceedings abstracts, simple summaries, posters, and editorial notes.
- **EC5:** Non-academic dissemination studies.

2.6 PICOS framework for study selection

To ensure transparency and methodological consistency in the study selection process, the inclusion criteria were structured according to the PICOS framework (Population, Intervention, Comparator, Outcomes, and Study Design). This framework is widely used in systematic reviews to clarify the scope of the evidence and the characteristics of the included studies.

In the context of this review, the population refers to learners and participants involved in educational contexts where climate fiction (cli-fi) or eco-fiction is used as part of teaching and learning processes. The intervention corresponds to pedagogical practices that incorporate climate fiction as a learning resource in reading, writing, literature, or climate education activities. Because the field is still emerging and many studies adopt exploratory or qualitative designs, a formal comparator is often absent; therefore, studies without control groups were also considered. The outcomes focus on educational effects related to critical reading, argumentative writing, climate literacy, student engagement, and ecological awareness. Finally, the study designs include empirical studies, qualitative case studies, theoretical analyses, and literature reviews addressing the pedagogical use of climate fiction.

Table 1 summarizes the PICOS criteria applied in the selection of studies.

In the reviewed literature, critical reading skills are generally conceptualized as the ability to interpret, analyze, and evaluate texts in relation to broader socio-environmental contexts. In studies addressing climate fiction, this competency is often associated with students' capacity to interpret narrative representations of climate change, identify ethical dilemmas, and critically examine the relationship between human and ecological systems.

Argumentative writing refers to the ability to construct reasoned positions supported by textual evidence and structured arguments. Within the analyzed studies, this outcome is typically reflected in activities such as argumentative essays, reflective writing tasks, position papers, or classroom debates about climate change and sustainability issues. However, the majority of studies evaluate argumentative development qualitatively through textual analysis or classroom observations rather than through standardized quantitative instruments.

2.7 Selection process

The filtering process was carried out in three phases:

1. Initial screening by title and abstract
2. Full-text review
3. Removal of inter-database duplicates

TABLE 1 PICOS framework used for study selection.

PICOS element	Description in this review
Population (P)	Students or learners participating in formal or non-formal educational contexts, including secondary and higher education settings where climate fiction or eco-fiction is used in teaching and learning processes.
Intervention (I)	Educational activities incorporating climate fiction (cli-fi) as a pedagogical resource, including critical reading, literary analysis, creative writing, classroom discussions, and climate change education projects.
Comparator (C)	In most studies no explicit comparator is present. The review therefore includes studies without control groups, given the exploratory nature of the field.
Outcomes (O)	Development of critical reading skills, argumentative writing abilities, climate literacy, ecological awareness, student engagement, and socio-environmental agency.
Study Design (S)	Empirical educational studies, qualitative case studies, theoretical analyses, conceptual papers, and literature reviews addressing cli-fi in educational contexts.

Inclusion decisions were recorded following the PRISMA scheme. Duplicate records between Scopus, WoS, and ERIC were identified manually and through exact title matching (See Figure 1).

2.8 Data extraction and systematization

For the included studies, the following variables were extracted:

- Author(s) and year
- Affiliation
- Keywords
- Scientific journal
- Number of citations

2.9 Methodological quality assessment

The methodological quality of the included studies was assessed using an *ad hoc* framework adapted to the heterogeneous nature of the corpus, which included theoretical analyses, qualitative case studies, and empirical educational interventions. Each study was examined according to: (a) clarity of research questions and objectives; (b) coherence between theoretical framework and methodological design; (c) transparency of data collection and analysis procedures (when applicable); (d) pedagogical relevance of the intervention or proposal; and (e) validity and clarity of reported educational outcomes. Although no studies were excluded on the basis of quality alone, this appraisal informed the weighting of evidence in the qualitative synthesis and the identification of methodological gaps within the field.

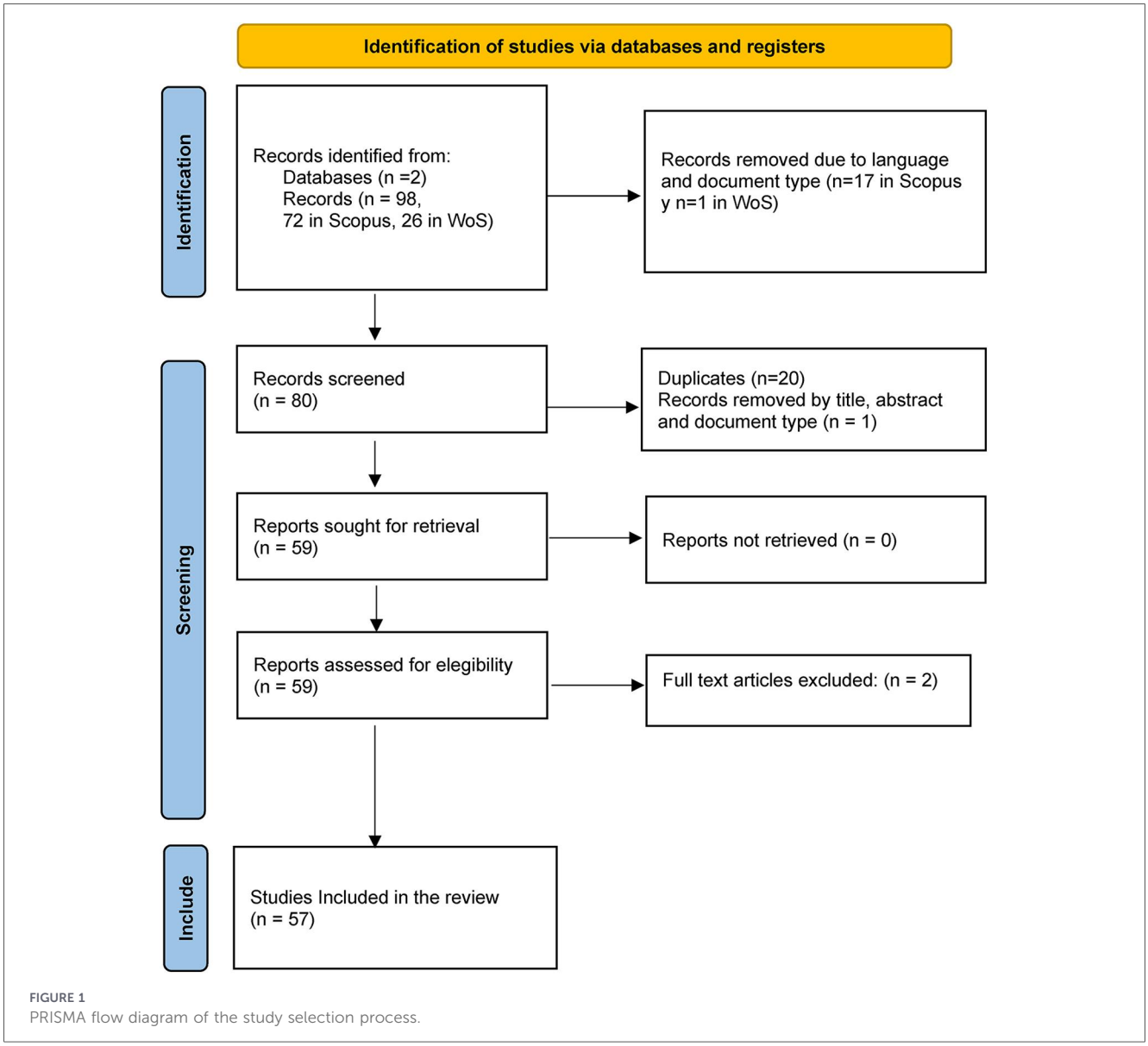


FIGURE 1
PRISMA flow diagram of the study selection process.

2.10 Synthesis strategy

A bibliometric analysis and a thematic qualitative synthesis were employed, oriented toward identifying patterns, pedagogical approaches, impacts on literacy, and research gaps.

2.11 Treatment of theoretical and conceptual studies

Given the interdisciplinary nature of the field, the corpus includes both empirical educational studies and theoretical or conceptual contributions from literary studies and ecocriticism. Empirical studies reporting classroom interventions or educational experiences were primarily used to identify documented educational outcomes. In contrast, theoretical works were considered to provide conceptual frameworks, pedagogical interpretations, and critical perspectives on the educational potential of climate fiction. Consequently, claims about measurable educational impact are derived mainly from empirical studies, while theoretical contributions inform the broader interpretation of the field.

3 Results

3.1 Temporal trends in productivity and citation

The temporal evolution of scientific output shows sustained growth between 2015 and 2025 (See Figure 2). During the first four years (2015–2018), publication levels were minimal, with 1 to 2 documents per year. From 2019 onward, a progressive increase is observed, reaching 5 publications in 2020, followed by 8 in 2021 and a slight decline in 2022 (4 publications). Subsequently, the volume rises again with 13 documents in 2024 and the period maximum in 2025 with 16 publications. This sequence reflects a gradual expansion of academic interest in the analyzed theme.

Regarding citations, more pronounced variability is observed throughout the period. Between 2015 and 2019, the number of citations remains at low levels (9 and 3 citations per year, respectively). In 2020, a notable increase occurs (56 citations), followed by a partial decline in 2021 (32 citations), and a new rise in 2022 (46 citations). The absolute peak appears in 2023 with 79 citations. Thereafter, the indicator decreases in 2024 (19 citations) and stands at 0 in 2025, a pattern expected for very recent years due to the natural lag in citation processes. Overall, productivity and citations exhibit asynchronous behavior, although both indicate a general increase in academic interest during the analyzed period.

3.2 Most influential journals

Table 2 presents the ten journals with the greatest influence within the analyzed corpus, considering both the number of documents published and accumulated citations. Among them, *Text* (Australia) ranks as the journal with the highest productivity, with four documents and eight citations. It is followed by *Humanities* (Switzerland) with three documents and a total of 42 citations, reflecting higher relative citation performance compared to publication volume.

At an intermediate level appear *Nature, Culture and Literature*, *Green Letters*, and *Australian Journal of Environmental Education*, each with two documents. However, their citation impact is uneven: while the first two register four and two citations respectively, the third shows no citations during the period considered.

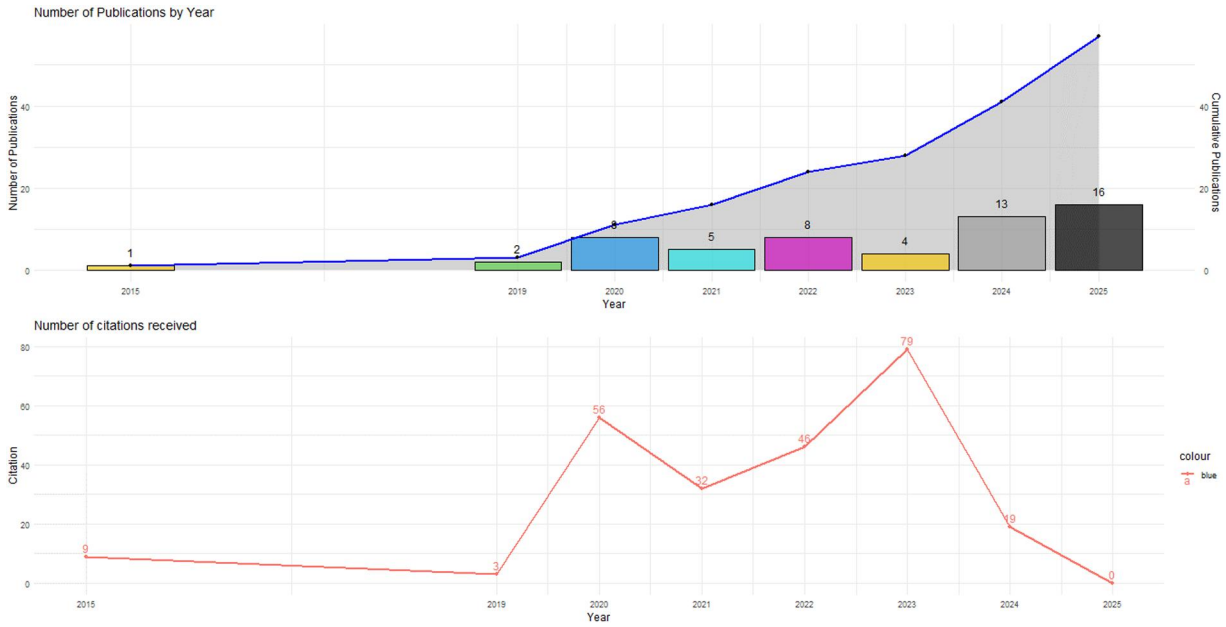


FIGURE 2
Annual trends in scientific publications and citations (2015–2025).

The group of journals with a single document displays higher citation patterns in some cases. *Environmental Communication* stands out as the journal with the highest number of citations (74), followed by *Curriculum Inquiry* (29 citations) and *Environmental Education Research* (26 citations). Finally, *Journal of Adolescent and Adult Literacy* and *Kamchatka* present moderate citation levels, with nine citations each.

Taken together, these ten journals constitute the most relevant editorial core for the scientific development of cli-fi in educational contexts, combining concentration of publications and citation impact.

TABLE 2 Most productive and highly cited journals in the analyzed corpus (number of documents and total citations).

Journal	Documents	Citations
Text (Australia)	4	8
Humanities (Switzerland)	3	42
Nature, Culture and Literature	2	4
Green Letters	2	2
Australian Journal of Environmental Education	2	0
Environmental Communication	1	74
Curriculum Inquir	1	29
Environmental Education Research	1	26
Journal of Adolescent and Adult Literacy	1	9
Kamchatka	1	9

3.3 Leading countries

Table 3 shows the countries included in the co-authorship analysis according to number of documents and citations. The United States presents the highest volume of publications (15 documents), followed by Australia (10 documents). In a second group are Sweden and the United Kingdom, both with four documents, while Austria, Belgium, and India register three documents each. Turkey and the Netherlands contribute two documents each, and the remaining countries present a single document.

In terms of citations, asymmetrical patterns are observed with respect to productivity. Spain records the highest number of citations (74), followed by Austria (48), Turkey (27), Finland (24), and the Netherlands (15). Taiwan and Norway each present nine citations, while the United States accumulates only three citations and Australia one citation, despite having the highest productivity levels. Other countries such as Belgium, India, Sweden, and the United Kingdom register no citations during the period analyzed.

Overall, the analysis reveals that international co-authorship presents a heterogeneous distribution both in publication volume and in citation, with no direct correspondence between the two indicators.

TABLE 3 Countries with the highest publication output and citation counts in cli-fi and education.

Country	Documents	Citations
United States	15	3
Australia	10	1
Sweden	4	0
United Kingdom	4	0
Austria	3	48
Belgium	3	0
India	3	0
Netherlands	2	15
Turkey	2	27
Finland	1	24
France	1	1
Germany	1	0
Ireland	1	2
Norway	1	9
Singapore	1	1
Spain	1	74
Taiwan	1	9

3.4 Thematic areas

Figure 3 shows the keyword co-occurrence map generated with VOSviewer, in which multiple interconnected nodes are identified. The keywords with the largest node size correspond to *climate fiction, ecocriticism*, and *climate change*, indicating higher frequency of appearance in the corpus. Around these central terms, different groupings of keywords related to environmental, literary, and educational approaches can be observed.

The map displays various connections between terms associated with literary perspectives (*cli-fi, ambient literature, science fiction, creative writing*), as well as concepts from the environmental domain (*anthropocene, ecology, climate models, environmental justice*). Keywords linked to educational processes also appear, such as *climate change education* and *ESD*. The lines connecting the nodes reflect co-occurrence relationships, while the color assigned by the software distinguishes clusters of terms with greater thematic proximity.

Taken together, the map evidences a heterogeneous network in which elements of literature, environmental studies, and education converge, suggesting that the field presents a multidimensional articulation of concepts related to climate fiction and its academic treatment.

Beyond describing the thematic clusters identified in the keyword co-occurrence network, the bibliometric patterns reveal important implications for the conceptual positioning of cli-fi within educational research. The prominence of keywords related to ecocriticism, environmental humanities, and climate change indicates that the field remains strongly anchored in literary and cultural analysis rather than in pedagogical or curricular frameworks. Terms associated with educational practice, such as classroom, literacy, or writing, appear less

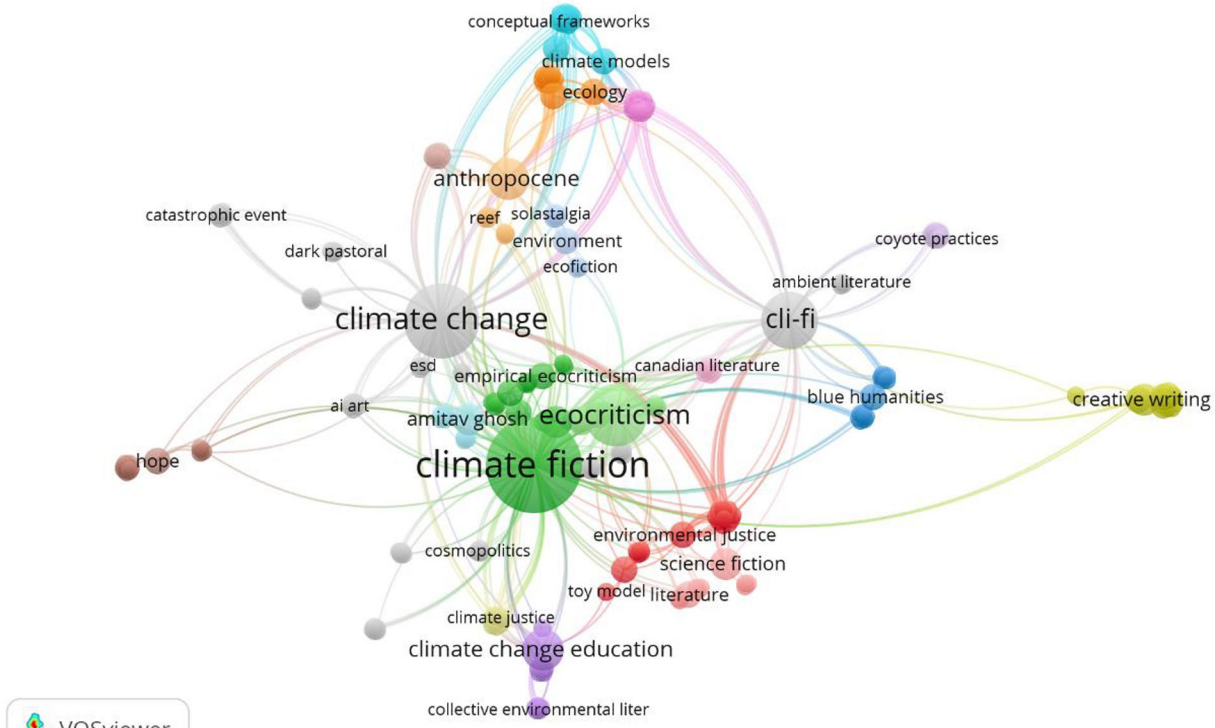


FIGURE 3
Keyword co-occurrence map of the literature on cli-fi and education (VOSviewer analysis).

central in the network, suggesting that the pedagogical dimension of cli-fi is still developing within the literature.

From a curricular perspective, this configuration indicates that cli-fi has been primarily conceptualized as an interpretative and affective resource rather than as a structured pedagogical tool for developing specific competencies such as argumentative writing or critical literacy. Consequently, the thematic distribution identified through the bibliometric analysis points to a gap between literary-ecocritical scholarship and classroom-oriented research. Addressing this gap represents a significant opportunity for integrating climate fiction more systematically into Language and Literature curricula, particularly in relation to argumentative discourse, climate literacy, and socio-environmental debate.

4 Systematic literature review

The final sample consisted of 57 publications on climate fiction and cli-fi in relation to the climate crisis, ecocriticism, and, to varying degrees, educational processes. The period spans 2015 to 2025, with a very marked increase starting in 2020, reflecting the consolidation of the field in the last five years. Of the total, approximately half of the works focus primarily on theoretical-critical contributions to literature and ecocriticism, while roughly 27 studies explicitly incorporate educational contexts, didactic proposals, or analyses of classroom experiences, particularly in secondary and higher education. From this corpus, the synthesis was organized around four axes: (a) approaches and educational contexts of cli-fi, (b) pedagogical strategies and uses of cli-fi, (c) reported educational outcomes, and (d) gaps and opportunities for research.

4.1 Approaches and educational contexts of cli-fi

The earliest works in the analyzed period already raise the need to integrate climate literature into reading and writing instruction as a way to reflect on the planet's future. Beach (2015), for example, proposes using literary texts to imagine possible futures and work on student agency in the face of the climate crisis, opening a line that would later be taken up by empirical studies and more developed curricular proposals. During the years prior to 2020, analyses of specific works or subgenres—such as eco-gothic or dystopian fiction—predominate, with emphasis on critical interpretation (Boast 2020; Lovell et al. 2019) rather than systematic didactic applications.

From 2020 onward, and especially between 2022 and 2025, an expansion of works that explicitly articulate cli-fi with formal educational contexts can be observed. In higher education, several studies describe experiences in university courses on literature, environmental humanities, or creative writing, where the reading of novels, short stories, and speculative narratives is combined with critical analysis and textual production activities (Bush 2024; Fetherston 2025; Webb et al. 2021). Other works explore programs or subjects in climate change education that incorporate cli-fi narratives as triggers for projects, debates, and

guided research activities (Diaz-Clark et al., 2025; Reynante et al., 2024, 2025).

In the school education domain, the evidence is more limited but significant. Lindgren Leavenworth and Manni (2021) analyze how young students interpret climate narratives and with what ideas about the future they articulate their readings. Rydsjö and Hval (2025) work with climate fiction for young readers from an English as a Foreign Language (EFL) perspective, focusing on intergenerational conflicts and emotions associated with the climate crisis. McLean Davies et al. (2024) examine how secondary English teachers address Indigenous climate fiction in the Australian context, highlighting tensions between decolonizing approaches and standardized curricula.

The corpus also includes works that, while not describing specific classroom interventions, discuss conceptual frameworks for thinking about the relationship between cli-fi, education, and environmental citizenship. Östman et al. (2025) propose a framework for analyzing aesthetic experiences and artistic expressions linked to ecological learning, while Lahtinen and Löytty (2024) problematize the limits of an "empirical ecocriticism" based on reader responses. These theoretical contributions function as a bridge between literary criticism and the design of pedagogical practices, positioning cli-fi as a privileged resource for exploring affects, imaginaries, and models of the future in educational contexts.

In summary, the reviewed studies show that the field has shifted from fundamentally literary and ecocritical analyses toward a broader spectrum in which cli-fi is integrated into language, literature, environmental humanities, science education, and cultural studies courses, with a clear concentration in higher education and a more reduced set of experiences at the secondary level.

4.2 Pedagogical strategies and approaches in the use of cli-fi

The pedagogical strategies identified in the studies are organized around three major approaches: (a) critical reading and guided discussion, (b) creative writing and speculative narrative, and (c) climate change education projects with emphasis on climate justice and sustainable futures.

In the first group, numerous works describe the use of climate fiction as a trigger for the critical analysis of discourses on climate change, the Anthropocene, and environmental justice. Fetherston (2025) proposes a "hyperobject reading" approach that invites students to trace how the climate crisis is represented across different genres—fantastic, science fiction, horror—and to evaluate its ethical and political implications. Studies focused on specific works and authors, such as Kingsolver, Wu Ming-Yi, or Amitav Ghosh, use close reading to work on notions of agency, vulnerability, and ecological responsibility (Gaffric, 2023; Giunta, 2022; Vaishnav, 2025). Although these works sometimes lie on the border between literary criticism and pedagogy, several explicitly state their potential to be transferred to literature and cultural studies classrooms.

The second approach groups experiences that combine cli-fi reading with creative writing activities, personal diaries, or the production of speculative stories. Webb et al. (2021, 2025) describe creative writing workshops in which university students work on trauma and climate change through the creation of narratives that link personal experiences with dystopian or post-apocalyptic scenarios. Bush (2024) analyzes a creative writing course centered on the climate crisis, where exercises in fabulation, myth rewriting, and character construction are used to explore emotions such as grief, hope, or guilt. Other studies focus on traditionally marginalized voices, for example, Berthiot's (2025) work on narratives by girls and adolescents that narrate trauma and resistance in contexts of ecological crisis, or Hankin and Hogarth's (2025) proposals for writing "with" non-human species from a posthumanist perspective.

The third group brings together proposals directly framed within climate change education and climate justice education. Reynante et al. (2024, 2025) develop learning experiences in which students design cli-fi stories and "engineering fiction" to imagine sustainable solutions, analyze the cognitive abstraction of the climate crisis, and work on a "pedagogy of hope" associated with solar and collective futures. Diaz-Clark et al. (2025) describe a course that moves "from science to solutions," combining scientific content with narrative activities, collaborative projects, and analysis of different ways of communicating climate change. Along this line, several studies incorporate elements of critical futures studies and climate justice, using cli-fi to question hegemonic narratives about progress, economic growth, and the distribution of risks (Rahamath Nisha and Horizan Prasanna Kumar, 2025; Reynante et al., 2025).

Finally, some works in secondary education explore specific classroom strategies. Rydsjö and Hval (2025) show how reading young adult cli-fi novels allows addressing intergenerational conflicts, feelings of expendability, and experiences of climatic precarity, while McLean Davies et al. (2024) analyze anti-colonial reading approaches to Indigenous climate fiction by secondary teachers, highlighting the importance of explicit ethical and political positionings in text selection and mediation.

Taken together, the pedagogical strategies range from relatively traditional textual discussions to more complex designs that integrate creative writing, future-oriented projects, climate justice, and decolonial pedagogies, evidencing a growing sophistication in the way cli-fi is mobilized as a didactic resource.

4.3 Reported educational outcomes

Regarding documented effects, most of the reviewed educational studies report results in three main dimensions: (a) student participation and engagement, (b) development of climate literacy and ecological awareness, and (c) construction of agency, hope, and critical positioning in the face of the climate crisis. Only a small subset focuses explicitly on argumentation and academic writing competencies.

Several works indicate that working with cli-fi increases students' affective and cognitive involvement. The experiences described by Webb et al. (2021, 2025), Bush (2024), and Berthiot (2025) underline that combining reading and creative writing enables students to articulate complex emotions linked to climate change—fear, anxiety, grief, but also hope—in such a way that they feel more motivated to participate in discussions

and share their texts. Convergingly, studies in climate change education indicate that elaborating their own stories and exploring alternative scenarios helps reduce the cognitive abstraction of the climate crisis and connect it to everyday and local contexts (Diaz-Clark et al., 2025; Reynante et al., 2024, 2025).

In terms of climate literacy and ecological awareness, articles working with young readers show that cli-fi can broaden understanding of the interrelationships between social and ecological systems. Lindgren Leavenworth and Manni (2021) observe that climate narratives influence how students imagine the future and perceive themselves as actors within crisis scenarios. Rydsjö and Hval (2025) report that reading cli-fi in EFL contexts opens spaces to discuss childhood vulnerability, generational inequalities, and power relations, while works centered on myths, legends, and local stories evidence potential for connecting traditional knowledge with Anthropocene imaginaries (Caracciolo, 2022; Tabur, 2023).

Results related to agency and hope appear recurrently in studies that articulate cli-fi with pedagogies of climate justice and critical futures studies. Reynante et al. (2025) highlight that "engineering fiction" projects enabled students to collectively imagine sustainable solutions, generating feelings of collective efficacy and commitment to climate action. Rahamath Nisha and Horizan Prasanna Kumar (2025) explicitly address climate-induced trauma and propose cli-fi as a device for working on fight, flight, freeze, and "fawn" responses ("4Fs"), suggesting that narrative elaboration can contribute to emotionally processing the crisis. Other studies discuss how cli-fi questions anthropocentric narratives and promotes posthumanist and multispecies perspectives, which in turn affect the way students conceive environmental justice and responsibility (Hankin and Hogarth, 2025; Kershen, 2025).

In contrast, fewer works systematically evaluate the impact on argumentation or academic writing competencies. Some articles analyze the construction of ethical and political arguments from climate narratives or examine how lecto-writing about the climate crisis can be structured in terms of claims, evidence, and counterarguments (Hennessy et al., 2024; Hoydis, 2025; Ralph, 2019). However, in most cases the evaluation is based on qualitative analysis of texts produced by students or descriptions of activities, rather than on quantitative measures of performance in argumentative writing, which limits the possibility of establishing robust causal relationships between the use of cli-fi and specific improvements in this competency.

In summary, the available evidence suggests that cli-fi and climate fiction, when integrated into reading and writing practices, favor student engagement, critical reflection on climate change, and the construction of alternative imaginaries. Nevertheless, results in terms of written argumentation remain incipient and require more systematic research designs.

4.4 Gaps, limitations, and opportunities for research

The systematic review also makes it possible to identify a series of gaps and limitations that open opportunities for future research. First, there is a clear geographical and linguistic bias. The majority of studies come from Anglophone contexts—United States, Australia, United Kingdom, and Nordic countries—and focus on literature written in English or translated into English. Only a reduced set of works explores cli-fi produced in other languages or situated in non-Western contexts, such as Turkish, Taiwanese, or Latin American literature (Gaffric, 2023; Irizarry, 2025; Tabur, 2023). This limits understanding of how the fictionalization of climate articulates with local histories, Indigenous cosmologies, and global inequalities.

Second, although nearly half of the corpus explicitly addresses educational contexts, descriptive studies, theoretical commentaries, and small-scale case studies predominate. Few articles specify experimental, quasi-experimental, or mixed-methods designs aimed at measuring changes in knowledge, attitudes, or competencies before and after cli-fi interventions; in the abstracts, only a small number mention surveys, experiments, or structured quantitative instruments. This makes it difficult to precisely evaluate the magnitude of the impact of these proposals on variables such as climate literacy, argumentative writing, or intention for climate action.

A third gap relates to the educational levels and disciplinary areas covered. Most experiences are situated in higher education, especially in humanities, literature, cultural studies, and, to a lesser extent, science education programs. Studies at the secondary level are less frequent and practically non-existent at the primary level, despite the considerable growth of children's and young adult climate literature. Likewise, the integration of cli-fi into Language and Literature courses with explicit emphasis on argumentative writing, rhetorical analysis, or multiliteracies remains marginal compared to proposals more focused on ethical and affective reflection.

Finally, most works focus on the classroom as the main setting and address only tangentially the relationship with educational policies, teacher training, and institutional conditions. Some exceptions discuss the role of national curricular frameworks or decolonial approaches in text selection (McLean Davies et al., 2024) or propose long-range conceptualizations about education for sustainability and aesthetic experiences (Östman et al., 2025), but in general there is a lack of studies that systematically connect cli-fi with curricular reforms, teacher professional development programs, or school initiatives for ecological transition.

These gaps open several promising lines: (a) developing rigorous empirical research that evaluates the effect of cli-fi-based reading and argumentative writing sequences on discursive competencies; (b) broadening the geographical focus toward the Global South and languages other than English, incorporating decolonial and plurilingual perspectives; (c) exploring the potential of cli-fi in primary education and non-formal contexts, such as reading clubs, community workshops, or interdisciplinary projects; and (d) articulating classroom work with climate education policies, teacher training, and institutional sustainability strategies.

Taken together, the systematic review shows that cli-fi has consolidated as a dynamic field at the intersection between literature, ecocriticism, and education, but also evidences that its potential for the development of reading and argumentative writing in Language and Literature remains underexploited and constitutes a fertile niche for future empirical research and innovative pedagogical proposals.

5 Discussion

The results evidence a sustained growth in scientific production on climate fiction in education between 2020 and 2025, especially in the university sphere. This increase is consistent with the rise of initiatives seeking to integrate the climate crisis into environmental humanities and literary education (Beach, 2015; Gaffric, 2023; Webb et al., 2021, 2025). In this context, cli-fi consolidates itself as a resource for articulating speculative futures, climate emotions, and socio-environmental agency, in line with the emphasis that environmental humanities have placed on climatic imagination and its affective dimensions (Berthiot, 2025; Bush, 2024).

The bibliometric findings also contribute to understanding how the field has evolved conceptually and pedagogically. The concentration of publications in journals linked to environmental humanities and literary studies suggests that cli-fi has been predominantly approached from interpretive and cultural perspectives. While this orientation has enriched theoretical debates on climate imagination and ecological narratives, it also explains why empirical research on literacy development and argumentative writing remains limited.

At the curricular level, these patterns indicate that the educational potential of cli-fi has not yet been fully translated into structured pedagogical frameworks. The relative absence of keywords associated with literacy assessment, writing instruction, or curricular design reflects the early stage of integration of cli-fi into formal educational programs. Therefore, the bibliometric evidence supports the need to move beyond descriptive literary analysis toward pedagogical research capable of evaluating how cli-fi can contribute to the development of discursive competencies and climate literacy in educational settings.

Beyond describing publication growth and citation patterns, the bibliometric trends reveal an uneven consolidation of cli-fi within educational research. While the steady increase in publications since 2020 reflects the growing centrality of climate narratives within environmental humanities and literary studies, the relatively low concentration of studies explicitly addressing argumentative writing indicates that this dimension remains peripheral within the current research landscape. The predominance of publications in humanities-oriented journals and the concentration of output in Anglophone contexts suggest that cli-fi has been primarily conceptualized as a cultural and affective resource, rather than as a structured pedagogical tool for developing discursive competencies. This imbalance points to a research gap in systematically exploring cli-fi as a didactic resource for argumentative literacy within formal Language and Literature curricula.

Nevertheless, this thematic expansion reveals clear imbalances. In geographical terms, research is concentrated in the United States, Australia, the United Kingdom, and Nordic countries, confirming the centrality of the Anglophone-Nordic axis in contemporary ecocriticism (McLean Davies et al., 2024; Östman et al., 2025). The scarce presence of the Global South and literatures in non-hegemonic languages—despite their relevance in terms of climatic vulnerability—reduces the cultural and epistemological diversity of the field, even though studies on Turkish, Taiwanese, or Latin American cli-fi (Gaffric, 2023; Irizarry, 2025; Tabur, 2023) demonstrate significant narrative potential.

From an educational perspective, cli-fi has been incorporated primarily in higher education and in humanistic disciplines such as literature, cultural studies, and creative writing (Bush, 2024; Webb et al., 2021). Its presence in secondary education is lower and practically non-existent in primary education (Lindgren Leavenworth and Manni, 2021; Rydsjö and Hval, 2025), indicating that the field has developed early pedagogical applications in university settings before expanding to earlier levels. The pedagogical strategies identified are organized around three orientations: critical reading and discussion, creative writing and speculative narrative, and climate change education projects oriented toward climate justice (Reynante et al., 2024, 2025). This diversity confirms cli-fi's capacity to integrate affective, epistemic, and projective dimensions of the climate crisis, ranging from trauma and grief to climatic hope and sustainable futures (Berthiot, 2025; Rahamath Nisha and Horizan Prasanna Kumar, 2025).

The documented learning outcomes are concentrated in climate literacy, systemic understanding, and ecological awareness (Lindgren Leavenworth and Manni, 2021; Rydsjö and Hval, 2025). Likewise, evidence of agency and student engagement is recorded, consistent with arguments that cli-fi reduces the cognitive abstraction of climate change and facilitates its contextualization in personal and collective experiences (Reynante et al., 2024, 2025). However, effects on discursive skills—particularly reading and argumentative writing—have been scarcely addressed. Existing studies analyze argumentative or rhetorical structures in a qualitative key (Cothren et al., 2023; Hennessy et al., 2024; Hoydis, 2025; Ralph, 2019), but do not implement systematic designs aimed at measuring improvements in written performance, thus identifying a relevant curricular gap, particularly in Language and Literature subjects.

Methodologically, a qualitative, theoretical orientation and reliance on small-scale case studies predominate, consistent with the interpretive tradition of literary criticism. The absence of experimental, quasi-experimental, or mixed-methods designs limits the evaluation of measurable educational effects and hinders the establishment of robust comparative relationships in the formative domain. Finally, cli-fi configures itself as a hybrid space where aesthetic, affective, and political dimensions converge, with three main thematic orientations: literary-ecocritical, pedagogical-creative, and educational-transformative. However, it has not yet fully consolidated as a curricular tool for the development of reading and writing competencies within the school system.

Taken together, the field finds itself in a phase of conceptual and pedagogical expansion, with advances at the affective and socio-environmental level, but with a methodological and curricular gap regarding the development of discursive and argumentative skills. Reducing geographical asymmetries, strengthening empirical designs, and articulating cli-fi with educational policies, teacher training, and curricula oriented toward climate education and sustainability constitute a key challenge for the coming years.

5.1 Theoretical implications

The findings suggest that cli-fi constitutes a fertile bridge between environmental humanities, ecocriticism, and literary studies. Its capacity to articulate climatic affects, imagined futures, and socio-environmental justice situates it within an aesthetic-educational regime in which narrative operates as mediation between experience and planetary crisis. The convergence of ecocritical approaches, pedagogies of futures, and posthumanist perspectives (Hankin and Hogarth, 2025; Kershen, 2025) points toward a redefinition of the role of literature in 21st-century climatic formation, shifting the emphasis from representation of the Anthropocene toward action and political imagination. This shift broadens the field of study beyond critical reading, incorporating agency, speculative design, and climatic affectivity as constitutive dimensions of learning.

5.2 Educational and curricular implications

The educational implications unfold on three levels. At the pedagogical level, cli-fi offers opportunities to integrate reading, discussion, and creative writing, incorporating complex affective experiences and reflective processes about climatic futures. At the disciplinary level, its incorporation into Language and Literature would make it possible to extend literacy beyond textual comprehension toward discursive and argumentative competencies, especially relevant in contemporary contexts of socio-environmental crisis. At the curricular level, cli-fi can act as an articulator between literary education and climate education, providing a pathway to transversalize sustainability and environmental justice without diluting linguistic or competency objectives.

At the curricular level, the integration of cli-fi into Language and Literature education offers concrete opportunities for designing learning sequences that combine critical reading with argumentative writing tasks focused on climate-related issues. For example, students may be guided to formulate argumentative essays or position papers based on cli-fi narratives, articulating claims about climate justice, sustainability, or socio-environmental responsibility supported by textual evidence and external sources. Such sequences would allow cli-fi to function not only as an affective and imaginative resource, but also as a scaffold for developing argumentative literacy aligned with curriculum standards in language education and education for sustainable development. This potential is documented, but not yet systematized.

5.3 Limitations and future research

The review presents three limitations. The first, inherent to the field, is the linguistic-geographical bias toward the Anglophone-Nordic axis. The second is the predominance of descriptive and qualitative studies, which limits the evaluation of comparable educational effects. The third is the scarcity of interventions centered on argumentative competencies, in contrast to what is observed in creative writing or futures studies.

From these limitations, several lines of future research are delineated:

(a) robust empirical designs (experimental, quasi-experimental, and mixed) oriented toward measuring effects on climate literacy, writing, and argumentation;
(b) geographical and linguistic expansion toward the Global South, Indigenous peoples, and non-hegemonic literatures;
(c) curricular integration in primary and secondary education, where cli-fi could play an early formative role;
(d) teacher training and educational policies, to connect classroom practices with institutional transformations;
(e) longitudinal research on climatic agency and imagined futures.

6 Conclusions

Climate fiction (cli-fi) has emerged in recent years as a relevant pedagogical resource at the intersection between literature, environmental humanities, and climate education. The results show a sustained expansion of scientific production between 2020 and 2025, with predominance in higher education and in humanistic disciplines. The pedagogical strategies identified—critical reading, creative writing, and futures-based learning—evidence a dynamic field oriented toward ecological awareness, student agency, and the construction of imagined futures. Nevertheless, these strategies are predominantly centered on affective and socio-environmental dimensions, while the development of discursive competencies, especially written argumentation, receives limited attention.

The documented educational experiences indicate that cli-fi facilitates the integration of climatic affects, critical perspectives, and ethical-political reflections, contributing to reducing the cognitive abstraction of climate change and to contextualizing it within personal and collective experiences. However, the absence of systematic empirical designs limits the possibility of establishing robust relationships between the use of cli-fi and improvements in academic competencies. Likewise, the geographical concentration of the field and the predominance of English as the language of research restrict the cultural and epistemological diversity of the educational debate.

In synthesis, the field finds itself in a phase of conceptual and pedagogical consolidation, yet it requires broadening its curricular and methodological scope in order to fully realize its formative potential. Cli-fi offers opportunities to enrich climate education from a literary perspective and, simultaneously, to strengthen literacy and argumentation, particularly in subjects such as Language and Literature. Exploring this potential constitutes both a challenge and an opportunity for future research and for the design of educational policies and practices situated within the context of global socio-environmental crisis.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

KQ–N: Writing – original draft, Investigation, Methodology, Formal analysis, Conceptualization. SP–M: Writing – review & editing, Conceptualization, Validation. DP–G: Methodology, Supervision, Data curation, Visualization, Validation, Software, Writing – review & editing.

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